Exhaustive search for matrix automorphisms

Thesis Progress Notes

Probabilistic Methods for Scalable Randomized Algorithms, Stochastic Systems, and Long-Term Theoretical Horizons

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Preliminary research statement for discussion with the MASDS Program Chair and prospective thesis advisor, where the central object is a stochastic process $\{X_t\}_{t\geq 0}$ on a state space $\mathcal{S}$, the target is a rigorous result of the form $\Pr(\textit{success}) \geq 1-\delta$, and the longer-term horizon includes questions such as $\mathrm{P} \stackrel{?}{=} \mathrm{NP}$ and $\zeta(s) = 0$.

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1 The Problem

Let n be even, and consider a deck of n cards split into two color classes: $n/2$ **red** cards and $n/2$ **green** cards. Cards of the same color are indistinguishable, so the state space is

$$\mathcal{S}_n = \{\sigma \in \{R, G\}^n : \#\{\text{red in } \sigma\} = n/2\} \qquad |\mathcal{S}_n| = \binom{n}{n/2}.$$

The **top-to-random shuffle** picks the top card and reinserts it uniformly at random into one of the n positions. On the fully identifiable deck this is the classical Markov chain on S_n with known spectrum and mixing behavior. On the colored deck we obtain a chain on $\mathcal{S}_n$; denote its transition matrix by

$$P_n = (P_n(\sigma, \tau))_{\sigma,\tau \in \mathcal{S}_n}.$$

1.1 The parent chain on S_n: known spectrum and mixing behavior

The phrase "classical Markov chain on S_n with known spectrum and mixing behavior" deserves unpacking, because the entire colored problem is defined by **comparison to this parent chain**.

Definition 1.1 (Fully identifiable deck)**.** *A deck of n cards is "fully identifiable" if every card is distinct. That is, we can tell card* 1 *from card* 2 *from ... from card n.*

A state of a "fully identifiable'' deck is then a *permutation* of $\{1, \dots, n\}$, i.e., an element of the symmetric group S_n, with $|S_n| = n!$. The top-to-random shuffle acts on S_n as a Markov chain with transition matrix

$$P^{\text{top}}(\sigma, \tau) = \frac{1}{n} \cdot \#\{\text{insertion positions that turn } \sigma \text{ into } \tau\}.$$

Known spectrum. The eigenvalues of P^{top} on S_n are *exactly known*:

$$\lambda_k^{(n)} = 1 - \frac{k}{n} \qquad k = 0, 1, \dots, n-1$$

with $\lambda_0 = 1$ (multiplicity 1). The multiplicity of $\lambda_k^{(n)}$ equals the dimension of the irreducible representation of S_n indexed by the partition $(n-k, 1^k)$.

The derivation, due to Diaconis–Fill–Pitman (1992), decomposes the group algebra $\mathbb{C}[S_n]$ into its irreducible components. Under the right regular representation,

$$\mathbb{C}[S_n] \;\cong\; \bigoplus_{\lambda \vdash n} (\dim V^\lambda) \cdot V^\lambda$$

where the sum runs over integer partitions λ of n and V^λ is the irreducible representation of S_n indexed by λ. The top-to-random operator T acts on $\mathbb{C}[S_n]$ by right multiplication by the element

$$\frac{1}{n}\sum_{j=1}^{n}\tau_j$$

where τ_j is the permutation "move the top card to position j." A self-contained sketch of why T acts as the scalar $1-k/n$ on $V^{(n-k,1^k)}$ is given in Appendix A; the full proof is in Diaconis–Fill–Pitman (1992). The multiplicity of $\lambda_k^{(n)} = 1 - k/n$ is therefore $\dim V^{(n-k,1^k)}$, given by the hook-length formula.

Known mixing behavior. The mixing time is $T_{\text{mix}}(\varepsilon) \asymp n \log n$, and the chain **has a cutoff** at time $n \log n$.

Consequences for the colored chain. The colored chain P_n arises as a **lumping** of the parent chain under the projection $\pi_n : S_n \to \mathcal{S}_n$ that forgets the identities of individual cards within each color class. Because the parent chain's spectrum and mixing behavior are fully characterized, three consequences follow for P_n.

1. **Predict the eigenvalues** of P_n: they must be a subset of $\{1 - k/n\}_{k=0}^{n-1}$, since lumpings preserve a subset of the parent spectrum.
2. **Predict the mixing time**: P_n should mix at the same $n \log n$ scale, with the same cutoff structure.
3. **Identify what is unknown**: which eigenvalues survive, with what multiplicities, and with what eigenvector structure.

1.2 Research questions

The overarching question posed by Professor Evans is to characterize the mixing properties, the eigenvalues, and the eigenvectors of P_n.

1. *Doubly stochasticity.* Is P_n doubly stochastic, and if so, what is the structural reason?
2. *Closed-form description.* Can the entries $P_n(\sigma, \tau)$ be described by an explicit formula?
3. *Spectrum and multiplicities.* What are the eigenvalues of P_n, with what multiplicities, and what are the corresponding eigenvectors?
4. *Mixing time.* Can the mixing time $T_{\text{mix}}(\varepsilon)$ of P_n be described exactly, including the leading constant?
5. *Cutoff.* Does P_n exhibit a cutoff phenomenon, and if so, at what time scale?
6. *Interpolation.* How do these properties vary as n grows, and in particular what is the behavior as one passes from the identifiable parent chain on S_n to the fully indistinguishable quotient on $\mathcal{S}_n$?

2 The $n = 4$ Worked Example

For $n = 4$ we have $n/2 = 2$ red and 2 green cards, so

$$|\mathcal{S}_4| = \binom{4}{2} = 6.$$

Enumerate the states as binary strings (R = 1, G = 0):

$$\text{RRGG}, \quad \text{RGRG}, \quad \text{RGGR}, \quad \text{GRRG}, \quad \text{GRGR}, \quad \text{GGRR}.$$

The transition matrix on these six states, as written on the whiteboard during the meeting, is

$P_4 =$

	RRGG ● ● ○○	RGRG ● ○ ●○	RGGR ● ○ ○●	GRRG ○ ● ●○	GRGR ○ ● ○●	GGRR ○ ○ ●●
RRGG ● ● ○○	$1/4 + 1/4$	$1/4$	$1/4$	0	0	0
RGRG ● ○ ●○	0	$1/4$	0	$1/4$	0	$1/4 + 1/4$
RGGR ● ○ ○●	0	0	$1/4$	$1/4$	$1/4 + 1/4$	0
GRRG ○ ● ●○	0	$1/4$	$1/4 + 1/4$	$1/4$	0	0
GRGR ○ ● ○●	0	0	0	$1/4$	$1/4 + 1/4$	$1/4$
GGRR ○ ○ ●●	$1/4 + 1/4$	$1/4$	0	0	0	$1/4$

2.1 Doubly stochastic

Every row sums to 1: this is the definition of a stochastic matrix. But every **column** also sums to 1: P_4 is doubly stochastic. This is not an accident of $n = 4$.

3 Automorphisms and Equitable Partitions of P_4

This section investigates the exact symmetry structure of the six-state chain. The transition matrix is treated as a weighted, directed object: a permutation is an automorphism only if it preserves every transition probability.

Let

$$\mathcal{S}_4 = \{\text{RRGG}, \text{RGRG}, \text{RGGR}, \text{GRRG}, \text{GRGR}, \text{GGRR}\}$$

and write $P_4 = (P_{ij})_{i,j=1}^{6}$ with states ordered as above.

A permutation $\varphi : \mathcal{S}_4 \to \mathcal{S}_4$ is a **matrix automorphism** if

$$P_{ij} = P_{\varphi(i),\varphi(j)} \qquad \text{for all } i, j.$$

The automorphism group is